

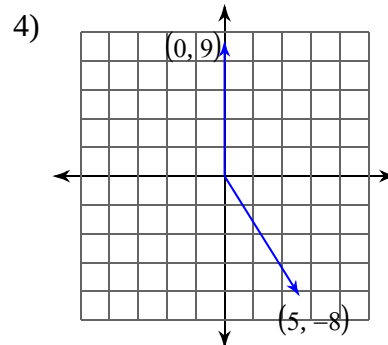
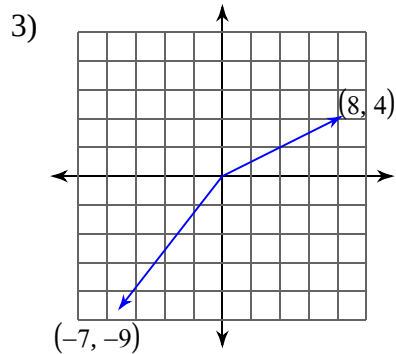
Two-Dimensional Vector Dot Products

Date _____ Period _____

Find the dot product of the given vectors.

1) $\vec{u} = \langle 3, 9 \rangle$
 $\vec{v} = \langle 6, 5 \rangle$

2) $\vec{u} = -\vec{i} + 5\vec{j}$
 $\vec{v} = -6\vec{i} - 2\vec{j}$

**State if the two vectors are parallel, orthogonal, or neither.**

5) $\vec{u} = \langle 4, -9 \rangle$
 $\vec{v} = \langle -9, 4 \rangle$

6) $\vec{u} = -5\vec{i} - 2\vec{j}$
 $\vec{v} = -10\vec{i} + 25\vec{j}$

Find the measure of the angle between the two vectors.